Table 1: Sorting-algorithm complexity.

Algorithm	**Best**	**Average**	**Worst**

Table 2: Numerical Integration Methods

Method	**Error**		**Order**
	Local	**Global**	
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Table 3: Standard derivatives and integrals by family.

Family	**Operation**		**Domain**
	Function	**Derivative**	
Power	x^n	nx^{n-1}	$n \in \mathbb{R}$
	$\frac{1}{x}$	$-\frac{1}{x^2}$	$x \neq 0$
	$\sqrt{x}$	$\frac{1}{2\sqrt{x}}$	$x > 0$
Trigonometric	$\sin x$	$\cos x$	$x \in \mathbb{R}$
	$\tan x$	$\sec^2 x$	$x \neq \frac{\pi}{2}$
Exponential	e^{ax}	ae^{ax}	$a \in \mathbb{R}$
	$\ln x$	$\frac{1}{x}$	$x > 0$
Chain rule $f(g(x))$		$f'(g(x))\, g(x)$	differentiable

Table 4: Semconductor Material Parameters

Material	**Electrical**		**Derived**	
	E_g (eV)	μ_n	$\sigma = \frac{1}{\rho}$	$\beta = \sqrt{\mu_n}$
Silicon	1.12	1.35×10^3	4.35×10^{-4}	36.7
Mean Gap			1.14 eV	

Table 5: Comparison of transform methods.

Transform	**Kernel function**	**Typical application**
Laplace	e^{-st}	transient circuits, control systems
Fourier	$e^{-j\omega t}$	spectra, signal filtering, madness
Z	z^{-n}	discrete systems, digital filters
All three convert differential relations into algebraic ones.		

Table 6: Boundary conditions and their meaning.

Name	**Form**	**Physical interpretation**
Dirichlet	$u = g$ on $\partial\Omega$	The value of the solution is prescribed on the boundary, for example a surface held at a fixed temperature.
Neumann	$\dfrac{\partial u}{\partial n} = g$	The flux across the boundary is prescribed; a zero value models a perfectly insulated wall.
Robin	$\alpha u + \beta \dfrac{\partial u}{\partial n} = g$	A weighted combination of the two above, used for convective heat exchange with a surrounding medium.

Table 7: Objects that must survive inside a cell.

Object	**Expression**	**Note**
Matrix	$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$	2×2 real
Determinant	$\begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1$	identity
Piecewise	$f(x) = \begin{cases} x^2 & x \geq 0 \\ -x & x < 0 \end{cases}$	continuous at 0
Integral	$\int_0^\infty e^{-st}\, dt = \frac{1}{s}$	$s > 0$
Limit	$\lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n = e$	classical
Nested index	$\sum_{\substack{i=1 \\ i \neq j}}^{n} a_{i_j}$	double subscript

Table 8: Network layer performance matrix.

Metric / Layer	**Training** (per epoch)		**Inference** (per sample)	
	Time	**Memory**	**Time**	**Memory**
Convolution	12.4 s	1.8 GB	3.1 ms	0.4 GB
Pooling	0.8 s	0.2 GB	0.2 ms	0.1 GB
Dense	4.6 s	3.4 GB	1.0 ms	0.9 GB
Total	17.8 s	5.4 GB	4.3 ms	1.4 GB